Please check the examination details below before entering your candidate information

Candidate surname	Other names

Centre Number | Candidate Number

Pearson Edexcel Level 3 GCE

Friday 16 May 2025

Afternoon

Paper reference **8FM0/26**

Further Mathematics

Advanced Subsidiary
Further Mathematics options
26: Further Mechanics 2
(Part of option J)

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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- Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a value of g is required, take $g = 9.8\text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
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Advice

- Read each question carefully before you start to answer it.
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- Check your answers if you have time at the end.

Turn over ▶

Pearson

1.

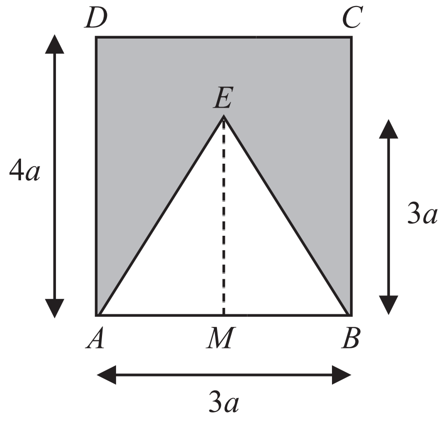

Figure 1

A uniform plane lamina, shown shaded in Figure 1, is formed by removing an isosceles triangle *ABE* from a rectangle *ABCD*.

- The midpoint of *AB* is *M*
- $AB = 3a$
- $AD = 4a$
- $EM = 3a$
- $AE = EB$

(a) Show that the distance of the centre of mass of the lamina from *AB* is $\frac{13a}{5}$

(5)

The lamina is suspended by a string attached to the lamina at *D*.

The lamina hangs freely in equilibrium.

(b) Find, to the nearest degree, the angle between *AD* and the vertical.

(3)

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Question 1 continued

Question 1 continued

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Question 1 continued

(Total for Question 1 is 8 marks)

2. A particle P moves on the x-axis. At time t seconds the velocity of P is v m s^{-1} in the positive x-direction, where

$$v = 3 - \sqrt{2t+1} \qquad t \geqslant 0$$

(a) Find the value of t when P is at instantaneous rest.

(2)

The acceleration of P at time t seconds is a m s^{-2} in the positive x-direction.

(b) Show that $a = \dfrac{1}{v-3}$

(3)

(c) Find the speed of P when it is decelerating at $\dfrac{4}{3}$ m s^{-2}

(2)

(d) Find the total distance travelled by P between $t = 0$ and $t = \dfrac{15}{2}$

[*Solutions relying on calculator technology are not acceptable.*]

(4)

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Question 2 continued

(Total for Question 2 is 11 marks)

P 7 4 0 7 4 A 0 7 1 6

3. [*In this question you may quote, without proof, the formula for the distance of the centre of mass of a circular arc from its centre.*]

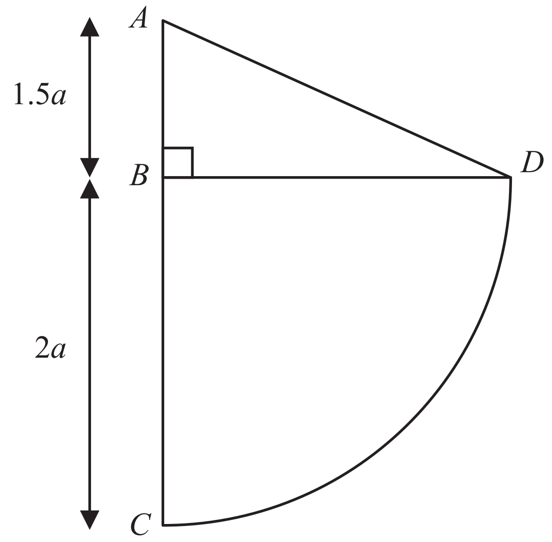

Figure 2

Uniform wire is used to form the rigid framework shown in Figure 2.

The framework consists of four straight pieces of wire, AB, BC, BD and AD and one piece in the shape of an arc of a circle, CD.

In the framework

- $AB = 1.5a$ and $BC = BD = 2a$
- CD is an arc of a circle of radius $2a$ and centre B
- ABC is a straight line and ABD is a right angle
- A, B, C and D all lie in the same plane

(a) Show that the distance of the centre of mass of arc CD from AC is $\dfrac{4a}{\pi}$

(2)

(b) Show that the distance of the centre of mass of the framework from AC is $\dfrac{17a}{2(8+\pi)}$

(4)

The framework is freely pivoted at A.

The framework is held in equilibrium, with BD horizontal, by an upward vertical force that is applied to the framework at D.

The mass of the framework is M.

The magnitude of the force exerted on the framework by the pivot at A is kMg.

(c) Find the value of k giving your answer in terms of π in its simplest form.

(3)

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Question 3 continued

Question 3 continued

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Question 3 continued

(Total for Question 3 is 9 marks)

P 7 4 0 7 4 A 0 1 1 1 6

4. A car is travelling round a circular track. The car moves with constant speed in a horizontal circle of radius r.

In an initial model,

- the car and driver are modelled as a single particle
- the track is modelled as being rough, so that there is sideways friction between the tyres of the car and the track with coefficient of friction μ
- the track is modelled as being **horizontal**

Using this model, the maximum speed at which the car can move round the circle of radius r without slipping sideways is $\frac{1}{2}\sqrt{gr}$.

(a) Show that $\mu = \frac{1}{4}$

(5)

In a refined model,

- the car and driver are modelled as a single particle
- the track is modelled as being rough, so that there is sideways friction between the tyres of the car and the track with coefficient of friction $\frac{1}{4}$
- the track is modelled as being **banked** at an angle θ to the horizontal

Using this model, the **minimum** speed at which the car can move round the circle of radius r without slipping sideways is $\sqrt{\frac{4rg}{35}}$

(b) Find the value of $\tan\theta$

(7)

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Question 4 continued

P 7 4 0 7 4 A 0 1 3 1 6

Question 4 continued

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Question 4 continued

Question 4 continued

(Total for Question 4 is 12 marks)

TOTAL FOR FURTHER MECHANICS 2 IS 40 MARKS

Please check the examination details below before entering your candidate information

Candidate surname	Other names

Centre Number | Candidate Number

Pearson Edexcel Level 3 GCE

Friday 17 May 2024

Afternoon

Paper reference **8FM0/26**

Further Mathematics

Advanced Subsidiary
Further Mathematics options
26: Further Mechanics 2
(Part of option J)

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

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- Check your answers if you have time at the end.

Turn over

Pearson

1.

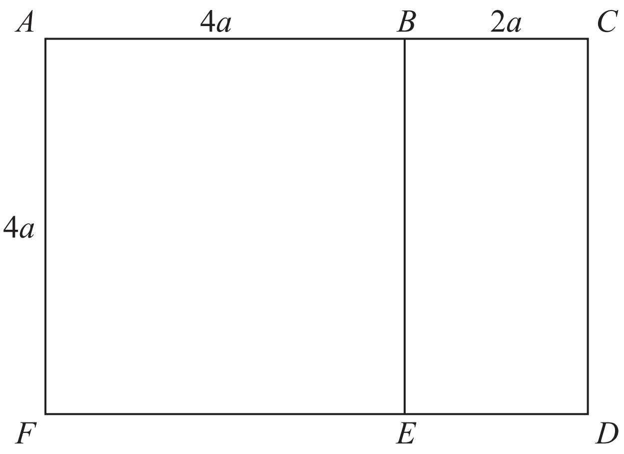

Figure 1

A uniform rod of length $24a$ is cut into seven pieces which are used to form the framework *ABCDEF* shown in Figure 1.

It is given that

- $AF = BE = CD = AB = FE = 4a$
- $BC = ED = 2a$
- the rods *AF*, *BE* and *CD* are parallel
- the rods *AB*, *BC*, *FE* and *ED* are parallel
- *AF* is perpendicular to *AB*
- the rods all lie in the same plane

The distance of the centre of mass of the framework from *AF* is d.

(a) Show that $d = \dfrac{19}{6}a$

(4)

(b) Find the distance of the centre of mass of the framework from *A*.

(3)

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Question 1 continued

(Total for Question 1 is 7 marks)

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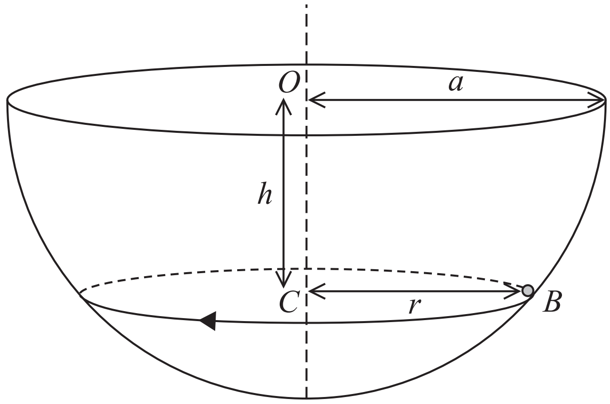

Figure 2

A thin hollow hemisphere, with centre O and radius a, is fixed with its axis vertical, as shown in Figure 2.

A small ball B of mass m moves in a horizontal circle on the inner surface of the hemisphere. The circle has centre C and radius r. The point C is vertically below O such that $OC = h$.

The ball moves with constant angular speed ω

The inner surface of the hemisphere is modelled as being smooth and B is modelled as a particle. Air resistance is modelled as being negligible.

(a) Show that $\omega^2 = \dfrac{g}{h}$

(6)

Given that the magnitude of the normal reaction between B and the surface of the hemisphere is $3mg$

(b) find ω in terms of g and a.

(3)

(c) State how, apart from ignoring air resistance, you have used the fact that B is modelled as a particle.

(1)

Question 2 continued

Question 2 continued

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Question 2 continued

(Total for Question 2 is 10 marks)

3. A particle P is moving along the x-axis. At time t seconds, P has velocity $v\,\text{m}\,\text{s}^{-1}$ in the positive x direction and acceleration $a\,\text{m}\,\text{s}^{-2}$ in the positive x direction.

In a model of the motion of P

$$a = 4 - 3v$$

When $t = 0,\ v = 0$

(a) Use integration to show that $v = k(1 - e^{-3t})$, where k is a constant to be found.

(7)

When $t = 0$, P is at the origin O

(b) Find, in terms of t only, the distance of P from O at time t seconds.

(4)

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P 7 5 6 7 6 A 0 8 1 6

Question 3 continued

Question 3 continued

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Question 3 continued

(Total for Question 3 is 11 marks)

4.

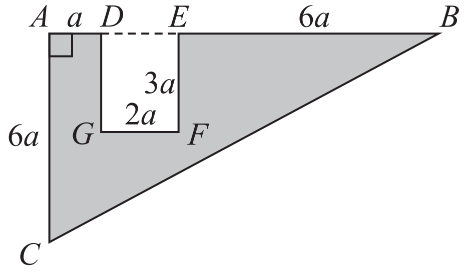

Figure 3

The uniform triangular lamina ABC has AB perpendicular to AC, $AB = 9a$ and $AC = 6a$. The point D on AB is such that $AD = a$.

The rectangle $DEFG$, with $DE = 2a$ and $EF = 3a$, is removed from the lamina to form the template shown shaded in Figure 3.

The distance of the centre of mass of the template from AC is d.

(a) Show that $d = \dfrac{23}{7}a$

(3)

The template is freely suspended from A and hangs in equilibrium with AB at an angle $\theta°$ to the downward vertical through A.

(b) Find the value of θ

(5)

A new piece, of exactly the same size and shape as the template, is cut from a lamina of a different uniform material. The template and the new piece are joined together to form the model shown in Figure 4. Both parts of the model lie in the same plane.

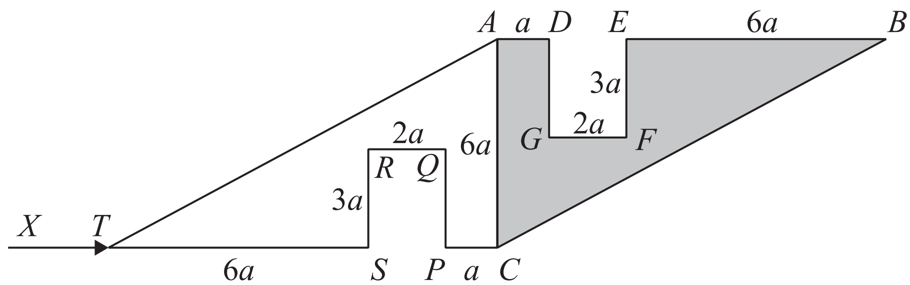

Figure 4

The weight of $CPQRSTA$ is W

The weight of $ADGFEBC$ is $4W$

The model is freely suspended from A.

A horizontal force of magnitude X, acting in the same vertical plane as the model, is now applied to the model at T so that AC is vertical, as shown in Figure 4.

(c) Find X in terms of W.

(4)

Question 4 continued

Question 4 continued

Question 4 continued

Question 4 continued

(Total for Question 4 is 12 marks)

TOTAL FOR FURTHER MECHANICS 2 IS 40 MARKS

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Please check the examination details below before entering your candidate information

Candidate surname	Other names

Centre Number | Candidate Number

Pearson Edexcel Level 3 GCE

Friday 19 May 2023

Afternoon

Paper reference **8FM0/26**

Further Mathematics
Advanced Subsidiary
Further Mathematics options
26: Further Mechanics 2
(Part of option J)

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

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Turn over ▶

1. Three particles of masses $4m$, $2m$ and km are placed at the points with coordinates $(-3,-1)$, $(6, 1)$ and $(-1, 5)$ respectively.

Given that the centre of mass of the three particles is at the point with coordinates $(\bar{x}, \bar{y})$

(a) show that $\bar{x} = \dfrac{-k}{k+6}$

(3)

(b) find $\bar{y}$ in terms of k.

(2)

Given that the centre of mass of the three particles lies on the line with equation $y = 2x + 3$

(c) find the value of k.

(2)

A fourth particle is placed at the point with coordinates $(\lambda, 4)$.

Given that the centre of mass of the four particles also lies on the line with equation $y = 2x + 3$

(d) find the value of λ.

(2)

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Question 1 continued

P 7 2 8 1 2 A 0 3 1 6

Question 1 continued

Question 1 continued

(Total for Question 1 is 9 marks)

2. A particle P is moving along the x-axis.
At time t seconds, $t \geqslant 0$, P has acceleration $a\,\mathrm{m\,s^{-2}}$ and velocity $v\,\mathrm{m\,s^{-1}}$ in the direction of x increasing, where

$$v = e^{2t} + 6e^{t} - kt$$

and k is a positive constant.

When $t = \ln 2$, $a = 0$

(a) Find the value of k.

(4)

When $t = 0$, the particle passes through the fixed point A.
When $t = \ln 2$, the particle is d metres from A.

(b) Showing all stages of your working, find the value of d correct to 2 significant figures.
[*Solutions relying entirely on calculator technology are not acceptable*.]

(4)

Question 2 continued

(Total for Question 2 is 8 marks)

3. A girl is cycling round a circular track.
The girl and her bicycle have a combined mass of 55 kg.
The coefficient of friction between the track surface and the tyres of the bicycle is μ.

The track is banked at an angle of 15° to the horizontal.

The girl and her bicycle are modelled as a particle moving in a horizontal circle of radius 50 m
The minimum speed at which the girl can cycle round this circle without slipping is $4.5\,\text{m}\,\text{s}^{-1}$

Using the model, find the value of μ.

(9)

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Question 3 continued

Question 3 continued

Question 3 continued

(Total for Question 3 is 9 marks)

4.

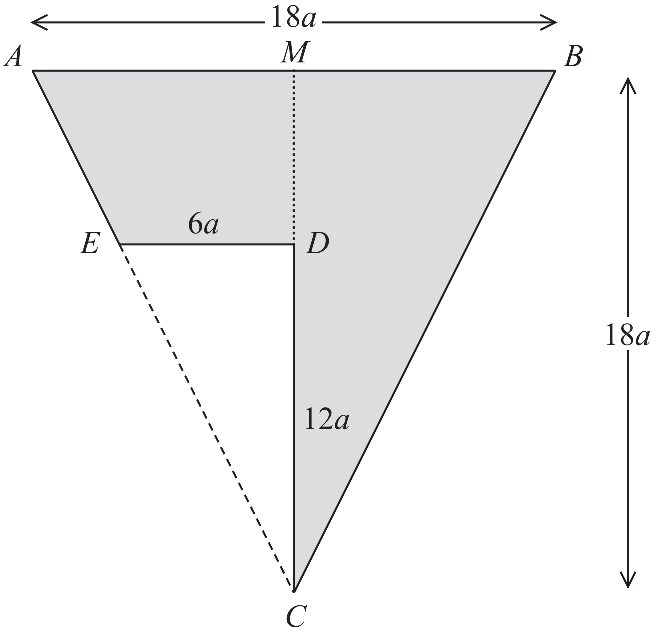

Figure 1

A uniform triangular lamina ABC is isosceles, with $AC = BC$. The midpoint of AB is M. The length of AB is $18a$ and the length of CM is $18a$.

The triangular lamina CDE, with $DE = 6a$ and $CD = 12a$, has ED parallel to AB and MDC is a straight line.

Triangle CDE is removed from triangle ABC to form the lamina L, shown shaded in Figure 1.

The distance of the centre of mass of L from MC is d.

(a) Show that $d = \dfrac{4}{7}a$

(4)

The lamina L is suspended by two light inextensible strings. One string is attached to L at A and the other string is attached to L at B.
The lamina hangs in equilibrium in a vertical plane with the strings vertical and AB horizontal.
The weight of L is W

(b) Find, in terms of W, the tension in the string attached to L at B

(3)

The string attached to L at B breaks, so that L is now suspended from A.
When L is hanging in equilibrium in a vertical plane, the angle between AB and the downward vertical through A is $\theta°$

(c) Find the value of θ

(7)

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Question 4 continued

Question 4 continued

Question 4 continued

P 7 2 8 1 2 A 0 1 5 1 6

Question 4 continued

(Total for Question 4 is 14 marks)

TOTAL FOR FURTHER MECHANICS 2 IS 40 MARKS

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Pearson Edexcel Level 3 GCE

Paper reference **8FM0/26**

Further Mathematics

Advanced Subsidiary
Further Mathematics options
26: Further Mechanics 2
(Part of option J)

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Pearson

1.

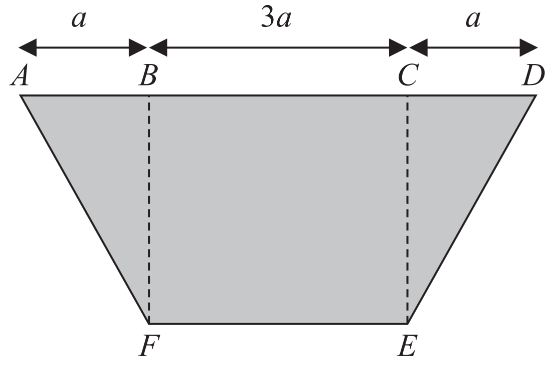

Figure 1

A uniform plane lamina is in the shape of an isosceles trapezium *ABCDEF*, as shown shaded in Figure 1.

- *BCEF* is a square
- $AB = CD = a$
- $BC = 3a$

(a) Show that the distance of the centre of mass of the lamina from *AD* is $\frac{11a}{8}$

(5)

The mass of the lamina is M

The lamina is suspended by two light vertical strings, one attached to the lamina at *A* and the other attached to the lamina at *F*

The lamina hangs freely in equilibrium, with *BF* horizontal.

(b) Find, in terms of M and g, the tension in the string attached at *A*

(2)

Question 1 continued

Question 1 continued

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Question 1 continued

(Total for Question 1 is 7 marks)

2.

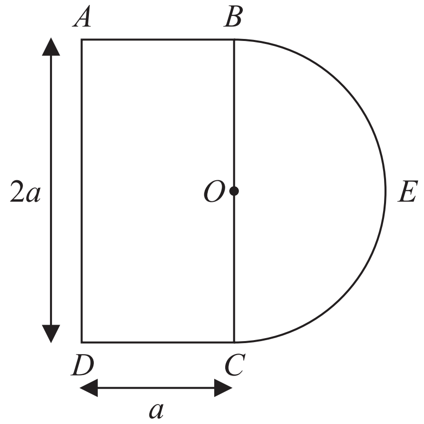

Figure 2

Uniform wire is used to form the framework shown in Figure 2.

In the framework

- $ABCD$ is a rectangle with $AD = 2a$ and $DC = a$
- BEC is a semicircular arc of radius a and centre O, where O lies on BC

The diameter of the semicircle is BC and the point E is such that OE is perpendicular to BC.

The points A, B, C, D and E all lie in the same plane.

(a) Show that the distance of the centre of mass of the framework from BC is

$$\frac{a}{6+\pi}$$

(5)

The framework is freely suspended from A and hangs in equilibrium with AE at an angle $\theta°$ to the downward vertical.

(b) Find the value of θ.

(4)

The mass of the framework is M.

A particle of mass kM is attached to the framework at B.

The centre of mass of the loaded framework lies on OA.

(c) Find the value of k.

(3)

Question 2 continued

Question 2 continued

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Question 2 continued

(Total for Question 2 is 12 marks)

3. A cyclist is travelling around a circular track which is banked at an angle α to the horizontal, where $\tan\alpha = \frac{3}{4}$

The cyclist moves with constant speed in a horizontal circle of radius r.

In an initial model,

- the cyclist and her cycle are modelled as a particle
- the track is modelled as being rough so that there is sideways friction between the tyres of the cycle and the track, with coefficient of friction μ, where $\mu < \frac{4}{3}$

Using this model, the maximum speed that the cyclist can travel around the track in a horizontal circle of radius r, without slipping sideways, is V.

(a) Show that $V = \sqrt{\frac{(3+4\mu)rg}{4-3\mu}}$

(7)

In a new simplified model,

- the cyclist and her cycle are modelled as a particle
- the motion is now modelled so that there is **no** sideways friction between the tyres of the cycle and the track

Using this new model, the speed that the cyclist can travel around the track in a horizontal circle of radius r, without slipping sideways, is U.

(b) Find U in terms of r and g.

(2)

(c) Show that $U < V$.

(2)

Question 3 continued

Question 3 continued

Question 3 continued

(Total for Question 3 is 11 marks)

Turn over

4. A particle P moves on the x-axis. At time t seconds the velocity of P is $v\,\mathrm{m\,s^{-1}}$ in the direction of x increasing, where

$$v = \frac{1}{2}\left(3e^{2t} - 1\right) \qquad t \geqslant 0$$

The acceleration of P at time t seconds is $a\,\mathrm{m\,s^{-2}}$

(a) Show that $a = 2v + 1$

(2)

(b) Find the acceleration of P when $t = 0$

(1)

(c) Find the exact distance travelled by P in accelerating from a speed of $1\,\mathrm{m\,s^{-1}}$ to a speed of $4\,\mathrm{m\,s^{-1}}$

(7)

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Question 4 continued

Question 4 continued

(Total for Question 4 is 10 marks)

TOTAL FOR FURTHER MECHANICS 2 IS 40 MARKS

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Pearson Edexcel Level 3 GCE

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Paper reference **8FM0/26**

Further Mathematics

Advanced Subsidiary
Further Mathematics options
26: Further Mechanics 2
(Part of option J)

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- Try to answer every question.
- Check your answers if you have time at the end.
- Good luck with your examination.

Turn over

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Pearson

1.

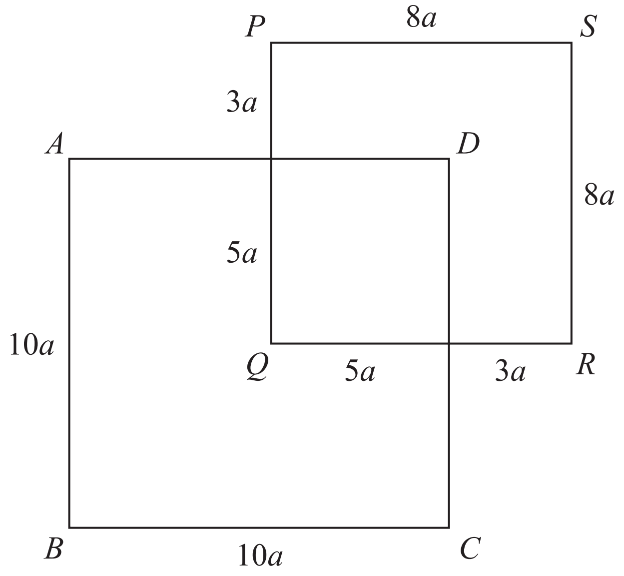

Figure 1

A uniform rod of length $72a$ is cut into pieces. The pieces are used to make two rigid squares, $ABCD$ and $PQRS$, with sides of length $10a$ and $8a$ respectively. The two squares are joined to form the rigid framework shown in Figure 1.

The squares both lie in the same plane with the rod AB parallel to the rod PQ.

Given that

- AD cuts PQ in the ratio $3:5$
- DC cuts QR in the ratio $5:3$

(a) explain why the centre of mass of square $ABCD$ is at Q.

(1)

(b) Find the distance of the centre of mass of the framework from B.

(5)

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Question 1 continued

(Total for Question 1 is 6 marks)

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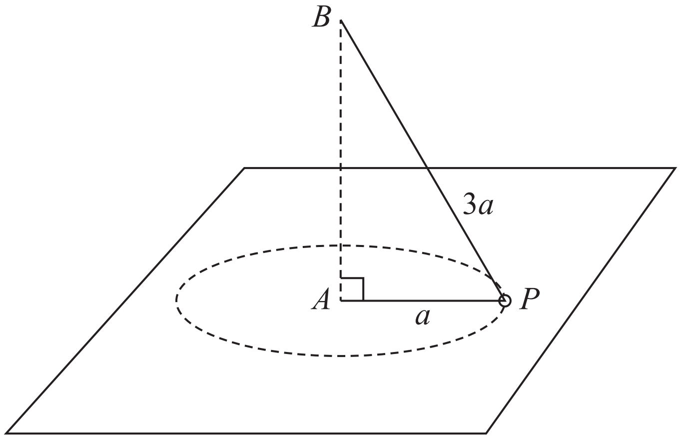

Figure 2

A small smooth ring P, of mass m, is threaded onto a light inextensible string of length $4a$.
One end of the string is attached to a fixed point A on a smooth horizontal table.
The other end of the string is attached to a fixed point B which is vertically above A.
The ring moves in a horizontal circle with centre A and radius a, as shown in Figure 2.

The ring moves with constant angular speed $\sqrt{\frac{2g}{3a}}$ about AB.

The string remains taut throughout the motion.

(a) Find, in terms of m and g, the magnitude of the normal reaction between P and the table.

(6)

The angular speed of P is now gradually increased.

(b) Find, in terms of a and g, the angular speed of P at the instant when it loses contact with the table.

(3)

(c) Explain how you have used the fact that P is smooth.

(1)

Question 2 continued

Question 2 continued

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Question 2 continued

(Total for Question 2 is 10 marks)

3.

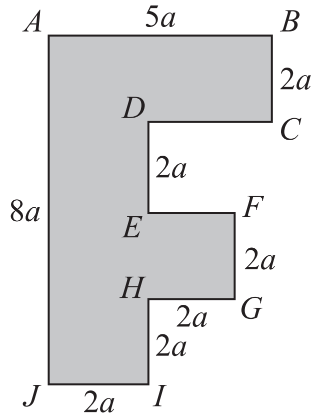

Figure 3

The uniform lamina *ABCDEFGHIJ* is shown in Figure 3.

The lamina has $AJ = 8a$, $AB = 5a$ and $BC = DE = EF = FG = GH = HI = IJ = 2a$.

All the corners are right angles.

(a) Show that the distance of the centre of mass of the lamina from *AJ* is $\dfrac{49}{26}a$

(5)

A light inextensible rope is attached to the lamina at *A* and another light inextensible rope is attached to the lamina at *B*. The lamina hangs in equilibrium with both ropes vertical and *AB* horizontal. The weight of the lamina is *W*.

(b) Find, in terms of *W*, the tension in the rope attached to the lamina at *B*.

(3)

The rope attached to *B* breaks and subsequently the lamina hangs freely in equilibrium, suspended from *A*.

(c) Find the size of the angle between *AJ* and the downward vertical.

(5)

Question 3 continued

Turn over

Question 3 continued

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Question 3 continued

(Total for Question 3 is 13 marks)

4. A particle P moves on the x-axis. At time t seconds, $t \geqslant 0$, P is x metres from the origin O and moving with velocity $v\,\mathrm{m\,s^{-1}}$ in the direction of x increasing, where

$$v = 5\sin 2t$$

When $t = 0$, $x = 1$ and P is at rest.

(a) Find the magnitude and direction of the acceleration of P at the instant when P is next at rest.

(4)

(b) Show that $1 \leqslant x \leqslant 6$

(4)

(c) Find the total time, in the first 4π seconds of the motion, for which P is more than 3 metres from O

(3)

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Question 4 continued

Question 4 continued

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Question 4 continued

Turn over

Question 4 continued

(Total for Question 4 is 11 marks)

TOTAL FOR FURTHER MECHANICS 2 IS 40 MARKS

Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel Level 3 GCE

Centre Number | Candidate Number

Thursday 08 October 2020

Afternoon | Paper Reference **8FM0/26**

Further Mathematics
Advanced Subsidiary
Further Mathematics options
26: Further Mechanics 2
(Part of option J)

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a value of g is required, take $g = 9.8\,\text{m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 3 questions.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

Pearson

1.

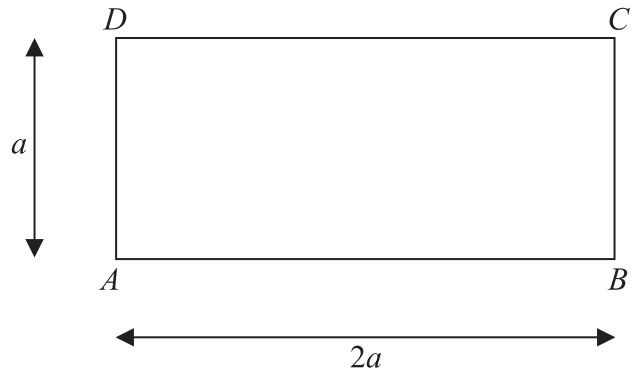

Figure 1

Figure 1 shows a uniform rectangular lamina $ABCD$ with $AB = 2a$ and $AD = a$
The mass of the lamina is $6m$.

A particle of mass $2m$ is attached to the lamina at A, a particle of mass m is attached to the lamina at B and a particle of mass $3m$ is attached to the lamina at D, to form a loaded lamina L of total mass $12m$.

(a) Write down the distance of the centre of mass of L from AB. You must give a reason for your answer.

(2)

(b) Show that the distance of the centre of mass of L from AD is $\dfrac{2a}{3}$

(3)

A particle of mass km is now also attached to L at D to form a new loaded lamina N.

(c) Show that the distance of the centre of mass of N from AB is $\dfrac{(k+6)a}{(k+12)}$

(4)

When N is freely suspended from A and is hanging in equilibrium, the side AB makes an angle α with the vertical, where $\tan\alpha = \dfrac{3}{2}$

(d) Find the value of k.

(6)

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Question 1 continued

Question 1 continued

Question 1 continued

(Total for Question 1 is 15 marks)

2.

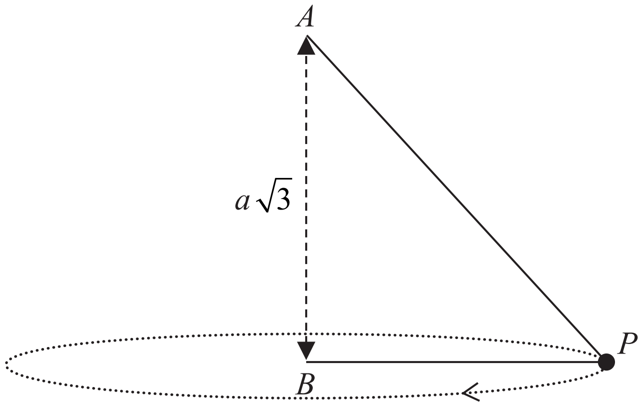

Figure 2

One end of a string of length $3a$ is attached to a point A and the other end is attached to a point B on a smooth horizontal table. The point B is vertically below A with $AB = a\sqrt{3}$
A small smooth bead, P, of mass m is threaded on to the string. The bead P moves on the table in a horizontal circle, with centre B, with constant speed U. Both portions, AP and BP, of the string are taut, as shown in Figure 2.

The string is modelled as being light and inextensible and the bead is modelled as a particle.

(a) Show that $AP = 2a$

(2)

(b) Find, in terms of m, U and a, the tension in the string.

(4)

(c) Show that $U^2 < ag\sqrt{3}$

(5)

(d) Describe what would happen if $U^2 > ag\sqrt{3}$

(1)

(e) State briefly how the tension in the string would be affected if the string were not modelled as being light.

(1)

Question 2 continued

Question 2 continued

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Question 2 continued

(Total for Question 2 is 13 marks)

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3. At time $t = 0$, a toy electric car is at rest at a fixed point O. The car then moves in a horizontal straight line so that at time t seconds ($t > 0$) after leaving O, the velocity of the car is $v\,\text{m s}^{-1}$ and the acceleration of the car is modelled as $(p + qv)\,\text{m s}^{-2}$, where p and q are constants.

When $t = 0$, the acceleration of the car is $3\,\text{m s}^{-2}$

When $t = T$, the acceleration of the car is $\frac{1}{2}\,\text{m s}^{-2}$ and $v = 4$

(a) Show that

$$8\frac{\mathrm{d}v}{\mathrm{d}t} = (24 - 5v)$$

(6)

(b) Find the exact value of T, simplifying your answer.

(6)

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Question 3 continued

Question 3 continued

(Total for Question 3 is 12 marks)

TOTAL FOR FURTHER MECHANICS 2 IS 40 MARKS

Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel Level 3 GCE

Centre Number | Candidate Number

Thursday 16 May 2019

Afternoon | Paper Reference **8FM0-26**

Further Mathematics

Advanced Subsidiary
Further Mathematics options
26: Further Mechanics 2
(Part of option J only)

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided – *there may be more space than you need*.
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a value of g is required, take $g = 9.8\ \text{m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question*.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

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Pearson

1.

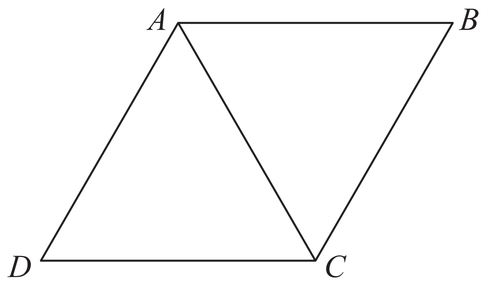

Figure 1

Five identical uniform rods are joined together to form the rigid framework *ABCD* shown in Figure 1. Each rod has weight *W* and length $4a$. The points *A*, *B*, *C* and *D* all lie in the same plane.

The centre of mass of the framework is at the point *G*.

(a) Explain why *G* is the midpoint of *AC*.

(1)

The framework is suspended from the ceiling by two vertical light inextensible strings. One string is attached to the framework at *A* and the other string is attached to the framework at *B*. The framework hangs freely in equilibrium with *AB* horizontal.

(b) Find

(i) the tension in the string attached at *A*,

(ii) the tension in the string attached at *B*.

(4)

A particle of weight kW is now attached to the framework at *D* and a particle of weight $2kW$ is now attached to the framework at *C*. The framework remains in equilibrium with *AB* horizontal and the strings vertical.

Either string will break if the tension in it exceeds $6W$.

(c) Find the greatest possible value of *k*.

(4)

Question 1 continued

Question 1 continued

Question 1 continued

(Total for Question 1 is 9 marks)

2. A car moves in a straight line along a horizontal road. The car is modelled as a particle. At time t seconds, where $t \geqslant 0$, the speed of the car is v m s^{-1}

At the instant when $t = 0$, the car passes through the point A with speed 2 m s^{-1}

The acceleration, a m s^{-2}, of the car is modelled by

$$a = \frac{4}{2+v}$$

in the direction of motion of the car.

(a) Use algebraic integration to show that $v = \sqrt{8t+16} - 2$

(6)

At the instant when the car passes through the point B, the speed of the car is 4 m s^{-1}

(b) Use algebraic integration to find the distance AB.

(6)

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Question 2 continued

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Question 2 continued

Question 2 continued

(Total for Question 2 is 12 marks)

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Turn over

3. A light inextensible string has length $8a$. One end of the string is attached to a fixed point A and the other end of the string is attached to a fixed point B, with A vertically above B and $AB = 4a$. A small ball of mass m is attached to a point P on the string, where $AP = 5a$.

The ball moves in a horizontal circle with constant speed v, with both AP and BP taut.

The string will break if the tension in it exceeds $\dfrac{3mg}{2}$

By modelling the ball as a particle and assuming the string does not break,

(a) show that $\dfrac{9ag}{4} < v^2 \leqslant \dfrac{27ag}{4}$

(7)

(b) find the least possible time needed for the ball to make one complete revolution.

(2)

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Question 3 continued

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Question 3 continued

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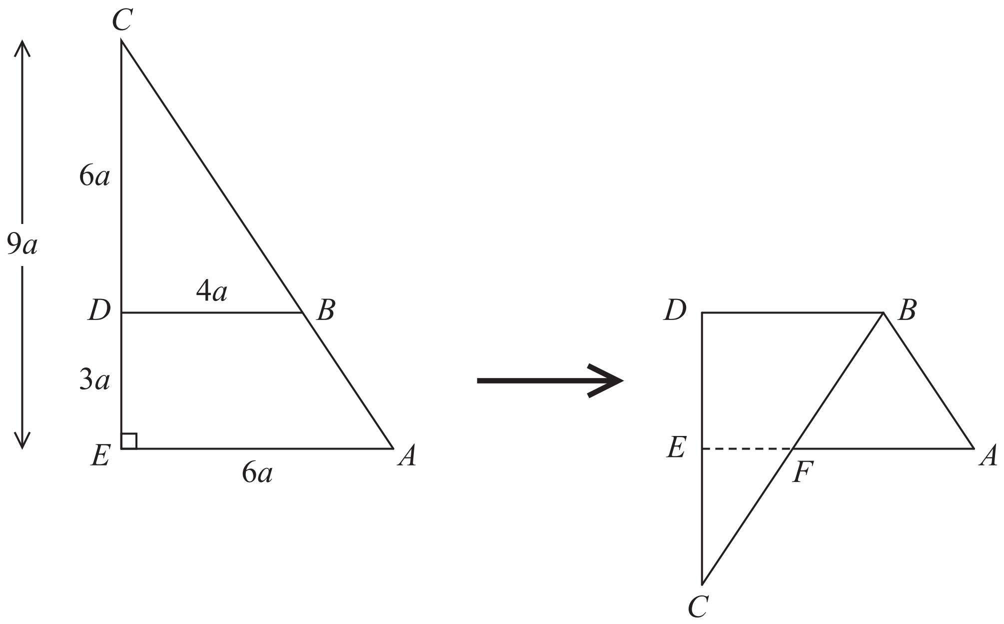
Question 3 continued

(Total for Question 3 is 9 marks)

4.



Triangular lamina   Folded lamina

Figure 2

The uniform triangular lamina $ABCDE$ is such that angle $CEA = 90°$, $CE = 9a$ and $EA = 6a$. The point D lies on CE, with $DE = 3a$. The point B on CA is such that DB is parallel to EA and $DB = 4a$. The triangular lamina is folded along the line DB to form the folded lamina $ABDECF$, as shown in Figure 2.

The distance of the centre of mass of the triangular lamina from DC is d_1

The distance of the centre of mass of the folded lamina from DC is d_2

(a) Explain why $d_1 = d_2$

(1)

The folded lamina is freely suspended from B and hangs in equilibrium with BA inclined at an angle α to the downward vertical through B.

(b) Find, to the nearest degree, the size of angle α.

(9)

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Question 4 continued

Question 4 continued

Question 4 continued

Question 4 continued

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Question 4 continued

Question 4 continued

(Total for Question 4 is 10 marks)

TOTAL FOR FURTHER MECHANICS 2 IS 40 MARKS

Write your name here

Surname	Other names

Pearson Edexcel Level 3 GCE

Centre Number	Candidate Number

Further Mathematics

Advanced Subsidiary
Further Mathematics options
26: Further Mechanics 2
(Part of option J only)

Thursday 17 May 2018 – Afternoon

Paper Reference
8FM0-26

You must have:
Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for each question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

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Unless otherwise indicated, whenever a numerical value of g is required, take $g = 9.8\,\mathrm{m\,s^{-2}}$ and give your answer to either 2 significant figures or 3 significant figures.

Answer ALL questions. Write your answers in the spaces provided.

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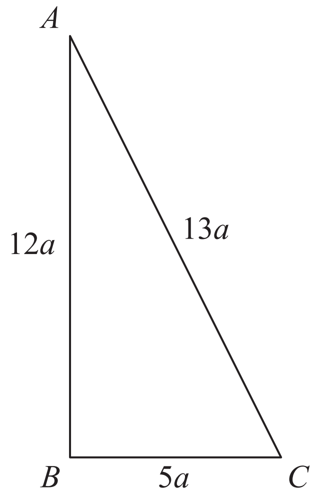

Figure 1

A thin uniform rod, of total length $30a$ and mass M, is bent to form a frame. The frame is in the shape of a triangle ABC, where $AB = 12a$, $BC = 5a$ and $CA = 13a$, as shown in Figure 1.

(a) Show that the centre of mass of the frame is $\frac{3}{2}a$ from AB.

(4)

The frame is freely suspended from A. A horizontal force of magnitude kMg, where k is a constant, is applied to the frame at B. The line of action of the force lies in the vertical plane containing the frame. The frame hangs in equilibrium with AB vertical.

(b) Find the value of k.

(3)

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Question 1 continued

(Total for Question 1 is 7 marks)

2. A car moves round a bend which is banked at a constant angle of $\theta°$ to the horizontal.

When the car is travelling at a constant speed of $80\,\text{km}\,\text{h}^{-1}$ there is no sideways frictional force on the car. The car is modelled as a particle moving in a horizontal circle of radius 500 m.

(a) Find the value of θ.

(7)

(b) Identify one limitation of this model.

(1)

The speed of the car is increased so that it is now travelling at a constant speed of $90\,\text{km}\,\text{h}^{-1}$
The car is still modelled as a particle moving in a horizontal circle of radius 500 m.

(c) Describe the extra force that will now be acting on the car, stating the direction of this force.

(1)

Question 2 continued

Question 2 continued

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Question 2 continued

(Total for Question 2 is 9 marks)

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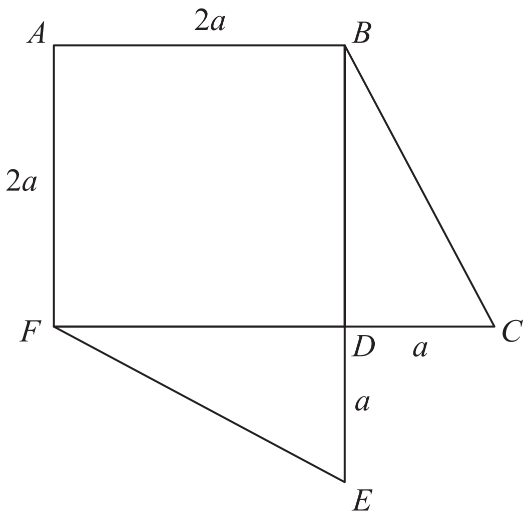

Figure 2

The lamina L, shown in Figure 2, consists of a uniform square lamina $ABDF$ and two uniform triangular laminas BDC and FDE. The square has sides of length $2a$. The two triangles are identical.

The straight lines BDE and FDC are perpendicular with $BD = DF = 2a$ and $DC = DE = a$.
The mass per unit of area of the square is M.
The mass per unit area of each triangle is $3M$.
The centre of mass of L is at the point G.

(a) Without doing any calculations, explain why G lies on AD.

(1)

(b) Show that the distance of G from D is $\dfrac{\sqrt{2}}{2}a$

(7)

The lamina L is freely suspended from B and hangs in equilibrium.

(c) Find the size of the angle between BE and the downward vertical.

(3)

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Question 3 continued

Question 3 continued

Question 3 continued

(Total for Question 3 is 11 marks)

4. A particle, P, moves on the x-axis. At time t seconds, $t \geqslant 0$, the velocity of P is $v\,\mathrm{m\,s^{-1}}$ in the direction of x increasing and the acceleration of P is $a\,\mathrm{m\,s^{-2}}$ in the direction of x increasing.

When $t = 0$ the particle is at rest at the origin O.

Given that $a = \dfrac{5}{2}(5 - v)$

(a) show that $v = 5(1 - \mathrm{e}^{-2.5t})$

(5)

(b) state the limiting value of v as t increases.

(1)

At the instant when $v = 2.5$, the particle is d metres from O.

(c) Show that $d = 2\ln 2 - 1$

(7)

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Question 4 continued

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Question 4 continued

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Question 4 continued

(Total for Question 4 is 13 marks)

TOTAL FOR FURTHER MECHANICS 2 IS 40 MARKS

Write your name here

Surname	Other names

Pearson Edexcel Level 3 GCE

Centre Number	Candidate Number

Further Mathematics

Advanced Subsidiary
Further Mathematics options
Paper 2J: Further Mechanics 1 and Further Mechanics 2

Sample Assessment Material for first teaching September 2017 **Time: 1 hour 40 minutes**	Paper Reference **8FM0/2J**

You must have: Mathematical Formulae and Statistical Tables, calculator	Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- There are **two** sections in this question paper. Answer **all** the questions in Section A and **all** the questions in Section B.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 80.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

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Pearson

SECTION B

Answer ALL questions. Write your answers in the spaces provided.

Unless otherwise indicated, whenever a numerical value of g is required, take $g = 9.8\,\mathrm{m\,s^{-2}}$ and give your answer to either 2 significant figures or 3 significant figures.

5. A particle P moves on the x-axis. At time t seconds the velocity of P is $v\,\mathrm{m\,s^{-1}}$ in the direction of x increasing, where

$$v = (t - 2)(3t - 10), \qquad t \geqslant 0$$

When $t = 0$, P is at the origin O.

(a) Find the acceleration of P at time t seconds.

(2)

(b) Find the total distance travelled by P in the first 2 seconds of its motion.

(3)

(c) Show that P never returns to O, explaining your reasoning.

(3)

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Question 5 continued

(Total for Question 5 is 8 marks)

6. A light inextensible string has length $7a$. One end of the string is attached to a fixed point A and the other end of the string is attached to a fixed point B, with A vertically above B and $AB = 5a$. A particle of mass m is attached to a point P on the string where $AP = 4a$. The particle moves in a horizontal circle with constant angular speed ω, with both AP and BP taut.

(a) Show that

(i) the tension in AP is $\dfrac{4m}{25}(9a\omega^2 + 5g)$

(ii) the tension in BP is $\dfrac{3m}{25}(16a\omega^2 - 5g)$.

(10)

The string will break if the tension in it reaches a magnitude of $4mg$.

The time for the particle to make one revolution is S.

(b) Show that

$$3\pi\sqrt{\frac{a}{5g}} < S < 8\pi\sqrt{\frac{a}{5g}}$$

(5)

(c) State how in your calculations you have used the assumption that the string is light.

(1)

Question 6 continued

Question 6 continued

(Total for Question 6 is 16 marks)

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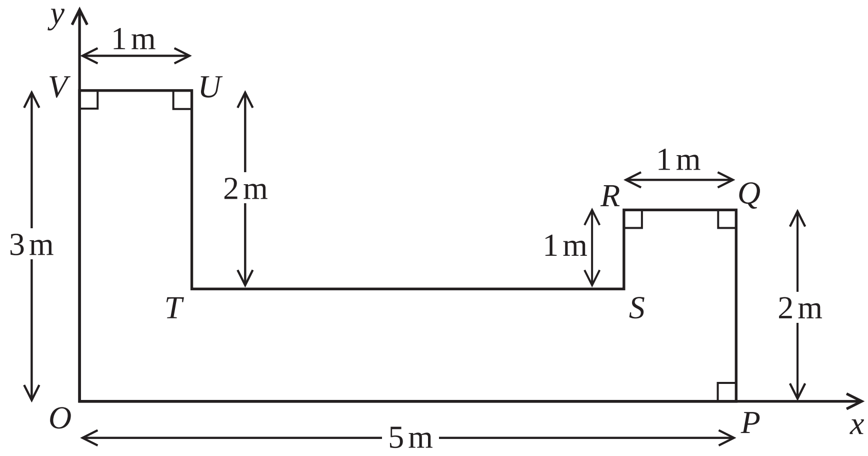

Figure 1

Figure 1 shows the shape and dimensions of a template *OPQRSTUV* made from thin uniform metal.
$OP = 5\,\text{m}$, $PQ = 2\,\text{m}$, $QR = 1\,\text{m}$, $RS = 1\,\text{m}$, $TU = 2\,\text{m}$, $UV = 1\,\text{m}$, $VO = 3\,\text{m}$.
Figure 1 also shows a coordinate system with O as origin and the x-axis and y-axis along OP and OV respectively. The unit of length on both axes is the metre.

The centre of mass of the template has coordinates $(\bar{x}, \bar{y})$.

(a) (i) Show that $\bar{y} = 1$

(ii) Find the value of $\bar{x}$.

(7)

A new design requires the template to have its centre of mass at the point (2.5, 1). In order to achieve this, two circular discs, each of radius r metres, are removed from the template which is shown in Figure 1, to form a new template L. The centre of the first disc is (0.5, 0.5) and the centre of the second disc is $(0.5, a)$ where a is a constant.

(b) Find the value of r.

(4)

(c) (i) Explain how symmetry can be used to find the value of a.

(ii) Find the value of a.

(2)

The template L is now freely suspended from the point U and hangs in equilibrium.

(d) Find the size of the angle between the line TU and the horizontal.

(3)

Question 7 continued

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Question 7 continued

Question 7 continued

(Total for Question 7 is 16 marks)

TOTAL FOR SECTION B IS 40 MARKS
TOTAL FOR PAPER IS 80 MARKS

Write your name here

Surname	Other names

Pearson Edexcel Level 3 GCE

Centre Number	Candidate Number

Specimen Paper

Paper Reference **8FM0-26**

Further Mathematics

Advanced Subsidiary

26: Further Mechanics 2

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 3 questions.
- The marks for each question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

Unless otherwise indicated, whenever a value of g is required, take $g = 9.8\,\mathrm{m\,s^{-2}}$ and give your answer to either 2 significant figures or 3 significant figures.

Answer ALL questions. Write your answers in the spaces provided.

1.

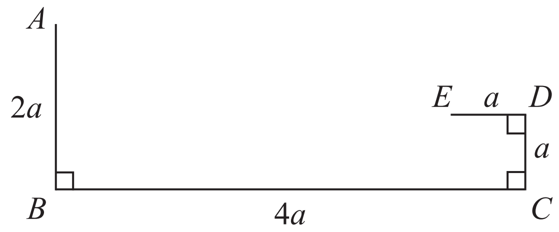

Figure 1

A thin uniform wire of length $8a$ is bent to form the framework $ABCDE$, where $AB = 2a$, $BC = 4a$, $CD = a$ and $DE = a$

AB is perpendicular to BC, BC is perpendicular to CD and CD is perpendicular to DE, as shown in Figure 1. The points A, B, C, D and E all lie in the same plane.

(a) Find the distance of the centre of mass of the framework from

(i) AB,

(ii) BC.

(7)

(b) Explain how you have used the fact that the wire is uniform in your solutions to part (a).

(1)

The framework is freely pivoted at A and hangs in equilibrium.

(c) Find the size of the angle between AB and the vertical.

(3)

The mass of the framework is M. A particle of mass kM is now attached to the framework at the midpoint of DE. The pivot at A is removed.

The loaded framework is now freely pivoted at the midpoint of BC and rests in equilibrium in a vertical plane with BC horizontal.

(d) Find the value of k.

(4)

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Question 1 continued

Question 1 continued

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Question 1 continued

(Total for Question 1 is 15 marks)

2. A circular race track is banked at an angle α to the horizontal where $\tan\alpha = \frac{3}{4}$

A car moves round the track at constant speed in a horizontal circle of radius 40 m. The car is modelled as a particle.

In an initial model of the motion of the car it is assumed that there is no sideways friction between the car and the track.

(a) Using this model, find the speed of the car.

(6)

The initial model of the motion of the car is now refined to make it more realistic. It is now assumed that there is sideways friction between the car and the track, with coefficient of friction μ. Under this refined model, the maximum speed at which the car can travel round the track in a horizontal circle of radius 40 m, is $39\,\mathrm{m\,s^{-1}}$.

(b) Using this refined model, find the value of μ

(10)

Question 2 continued

Question 2 continued

Question 2 continued

(Total for Question 2 is 16 marks)

3. A car moves in a straight line along a horizontal road.

At time t seconds, where $t \geqslant 0$, the speed of the car is $v\,\text{m s}^{-1}$.

The acceleration of the car, $a\,\text{m s}^{-2}$, is modelled by

$$a = \frac{50}{v} - \frac{v}{8}$$

Given that when $t = 0$, $v = 5$

(a) find v^2 in terms of t.

(7)

(b) Using your answer to part (a), find the limiting value of the speed of the car as t increases. You must give reasons for your answers.

(2)

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Question 3 continued

Question 3 continued

(Total for Question 3 is 9 marks)

TOTAL FOR PAPER IS 40 MARKS